

AHMED BODY PROJECT REPORT  
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**ABSTRACT**

In this report, we will the Ahmed body in ANSYS. The purpose of the lab is the simulate unsteady turbulent flow over the Ahmed body following the CFD processes by an interactive step by step approach. Also, we will compare our results with the experimentally results.

Firstly, we will create our geometry. we will make mesh our geometry with an unstructured construction. In the set-up section we will enter our boundary condition, reference value. We will choose our models. At the end, we will solve our problem to get results.

Keywords: Ahmed body, turbulent flow, CFD.

**NOMENCLATURE**

|        |                                        |
|--------|----------------------------------------|
| P      | Pressure                               |
| V      | Flow Velocity                          |
| T      | Temperature                            |
| d      | Diameter                               |
| $\rho$ | Density                                |
| $\mu$  | Dynamic viscosity                      |
| $F_D$  | Drag force                             |
| $C_D$  | Drag coefficient                       |
| A      | Characteristic frontal are of the body |
| $\nu$  | The kinematic viscosity.               |

**1. INTRODUCTION**

**1.1 Estimation of Drag and Lift on Ahmed Body Using CFD Analysis**

The aim of the paper is to evaluate the drag and lift coefficients of an Ahmed body.

The drag coefficient and lift coefficients of an automobile are found by changing the vehicle speed of the Ahmed body. The modeling is done in GAMBIT and the analysis in CFD. The aerodynamic analysis on Ahmed body showed a maximum drag coefficient of 1.146 and lift coefficient of 0.399. As the speed of Ahmed body increases the lift coefficient increased gradually and drag coefficient increased sharply and is highest at the top speed. At the vehicle speed increases, the drag coefficient increases which in turn increases the fuel consumption. The fuel consumption is the highest at the top speed. As the vehicle speed increases, the pressure in front of the vehicle has a maximum value of  $2.37 \times 10^3$  Pa. At the highest speed of the vehicle the velocity of air is maximum is at top side corners with a value of  $8.66 \times 10$  m/s. [1]

**1.2. The CFD (Computational Fluid Dynamics) analysis**

The CFD (Computational Fluid Dynamics) analysis is used to find the parameter in the automobile industry. This analysis is used to understand the fuel consumption, stability of the vehicle and passenger comfort. The air flow over ground vehicle is analyses and coefficient of drag is calculated using CFD (Ansys Fluent). For this calculation, Ahmed body (simplified car body) as ground vehicle is considered which is commonly used as test case in industry. The Ahmed body is made up of a round front part, a movable slant plane placed in the rear of the body to study the separation phenomena at 250, 350 angles, and a rectangular box, which connects the front and rear slant plane. The most significant feature of the body is the 250, 350 angle of rear slant. Air is used as a working fluid. The inlet velocity of fluid is 40 m/s. k- $\epsilon$  turbulent model used as a standard model. Two separate cases have been solved for two different front radius of Ahmed body R80, R120 and ground clearance at 20mm, 40mm and results are comparing. The results are present in the form of drag coefficient value and flow field which include velocity contour and velocity vector fields. the validation is carried out by simulation around the Ahmed body with the rear slant angle of 250, 350 the actual wind tunnel experimental data are compared with the results. [2]

**2. MATERIALS AND METHODS**

This figure shows the sketch that we will draw in the geometry model. The origin of the simulated is located to the reach of the Ahmed body. H is the height of the body. L is the length of the body.  $\theta$  (theta) is the slant angel. Uniform velocity is specified at inlet. Constant pressure will be given as a boundary condition in the outlet. The boundary top is symmetry. Other side boundary will be wall.

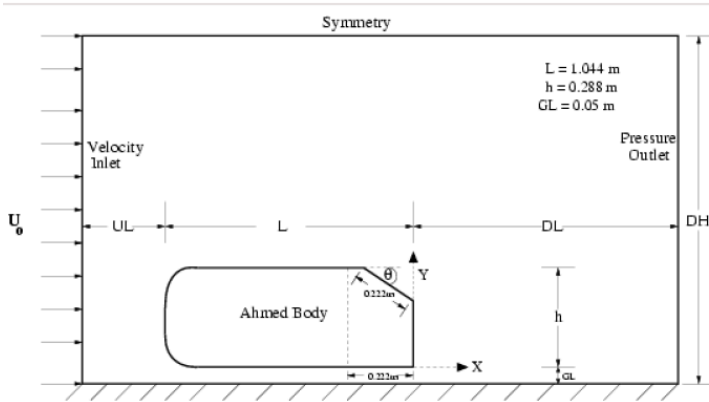


Figure: Ahmed body geometry

## 2.1. Formulas

Continuity equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Momentum equations:

For X-direction component:

$$\rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = - \frac{\partial p}{\partial x} + \mu \left[ \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right]$$

For Y-direction component:

$$\rho u \frac{\partial v}{\partial x} + \rho v \frac{\partial v}{\partial y} = - \frac{\partial p}{\partial y} + \mu \left[ \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right]$$

Reynold number:

$$Re = \frac{VL\rho}{\mu}$$

Drag coefficient:

$$F_D = C_D \rho v^2 A \frac{1}{2}$$

Skin coefficient ( $C_f$ ):

$$C_f = [2(Re) - 0.65]^{(-2.3)}$$

Wall shear stress ( $\tau_w$ )

$$\tau_w = C_f x \frac{1}{2} \rho v^2$$

## 2.2. General types of fluid flow

When calculating heat transfer or pressure loss, it is important to know whether the fluid flow is in transition, turbulent or laminar mode.

There are three general types of fluid flow:

### Laminar Flows

Conditions are low speed flow conditions of high viscosity fluids such as oil, laminar.

### Turbulent Flows

Processes At high speed flows, turbulent irregular fluids are also called turbulent flows. Variable, high-speed flow of low viscosity fluids such as air is called turbulent flow.

Flow that changes continuously between laminar and turbulent flow is also called transition flow.[3]

Accordingly, the flow:

$Re < 2300$  if laminar

Temporary if  $2,300 < Re < 4000$

$4000 < Re$  is turbulent flow.

Values:

Realizable k-epsilon model with non-equilibrium wall function for near wall treatment is used with Inlet velocity  $V = 40\text{m/s}$  and turbulence intensity is about 1% and turbulence viscosity ratio is 10 at inlet, turbulence intensity is about 5% and turbulence viscosity ratio is 10 at outlet. Density of air is  $1.225\text{ kg/m}^3$ , temperature is  $288.16\text{ K}$  and viscosity is  $1.7894\text{e-}05\text{ kg/m}^4$

**Reynold number:**

For airflow over the car:

$$Re = \frac{(1.225)(40\text{m/s})(0.288\text{m})}{(1.7894 \times 10^{-6})} = 788644.23$$

$788644.23 > 4000$

So, flow is turbulent

**Skin coefficient ( $C_f$ ):**

$$C_f = [2(Re) - 0.65]^{(-2.3)} = 3.9063 \times 10^{-3}$$

**Wall shear stress ( $\tau_w$ )**

$$\tau_w = C_f x \frac{1}{2} \rho v^2 = 3.8286\text{ Pa}$$

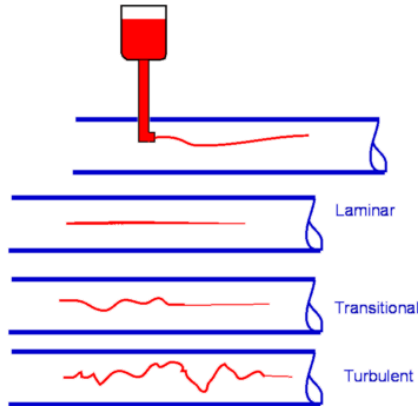


Figure: laminar, turbulent, transitive flow

of the geometry, especially Ahmed body. I use divisions type for sizing. In this way, I was able to increase the mesh size where I saw it critical. Finally, I set the names of all borders with the create name selection option.

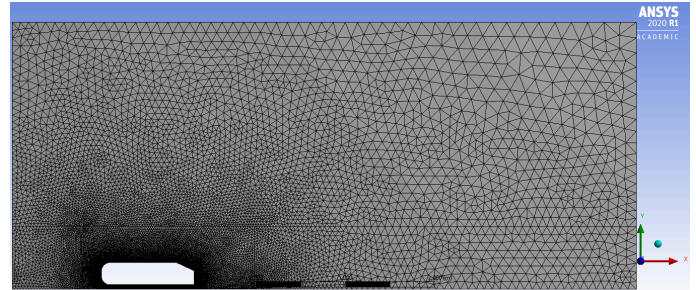


Figure: Ahmed body mesh generation

### 3. RESULTS AND DISCUSSION

Our problem is an unsteady turbulent over a simple car which is known as Ahmed body. It is a 2D geometry. Reynolds number is around 768,000 due to the inlet velocity and hydraulic diameter.

#### Create geometry:

To start geometry firstly, I will control the unit. It should be degree and meter. I start to draw geometry. After geometry drawing, I will create a face split for (dividing to 6 piece). Last part of geometry drawing, I use the merget method at top of the geometry for to join the pieces.

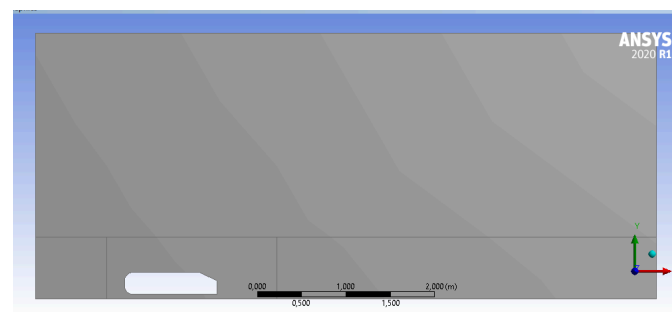


Figure: Ahmed body geometry in ANSYS

Comment: You can see merget and face split part in geometry.

#### Meshining:

At the meshing part, I use inflation method around the Ahmed body. This method put layers around the Ahmed body. Layer thickness is  $3.5 \times 10^{-7}$  m. after that, I use sizing all boundary

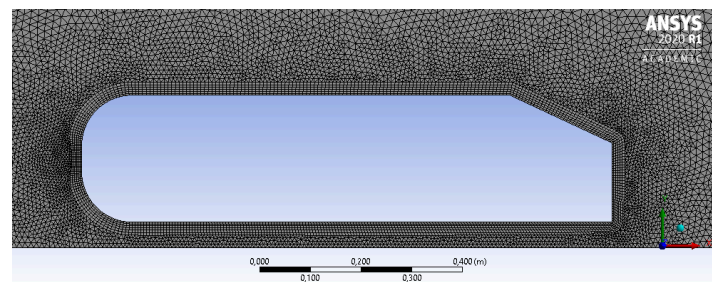


Figure: Ahmed body mesh generation xy plane zoom

Comment: The layers that I set with inflation are shown on the figure.

#### Set-up:

The problem will be transient analysis. Therefore, in the general part I will turn to time from steady to the transient. It is a turbulent flow so I will change my viscous model from laminar to the turbulent. I will use the standard k-epsilon model with nonequilibrium wall function. my material is air which have constant density and viscosity.

The boundary condition part, inlet will be velocity inlet and outlet will be pressure outlet. Other boundary conditions are symmetry and wall. Reference values is inlet. In the method part, I will change scheme from simple to the PISO. As it is high order model. Gradient is green- gauss cell based. Pressure is PRESTO. Other are QUICK. Initialization is inlet condition. And I will add drag coefficient fields in the report file than calculate.

Values:

Realizable k-epsilon model with non-equilibrium wall function for near wall treatment is used with Inlet velocity  $V = 40\text{m/s}$  and turbulence intensity is about 1% and turbulence viscosity ratio is 10 at inlet, turbulence intensity is about 5% and turbulence viscosity ratio is 10 at outlet. Density of air is  $1.225\text{ kg/m}^3$ , temperature is  $288.16\text{ K}$  and viscosity is  $1.7894\text{e-}05\text{ kg/m}^4$

Slant angle  $205^\circ$   $215^\circ$   $225^\circ$

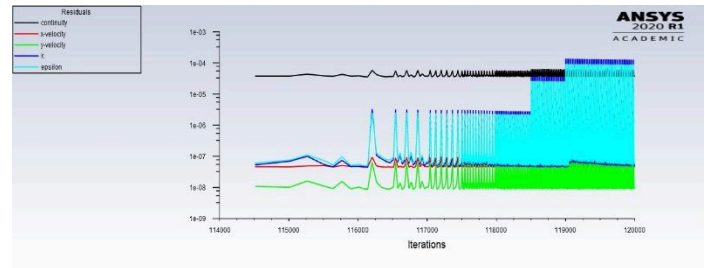
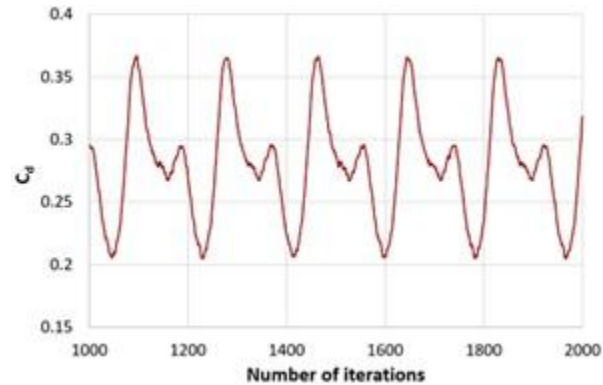
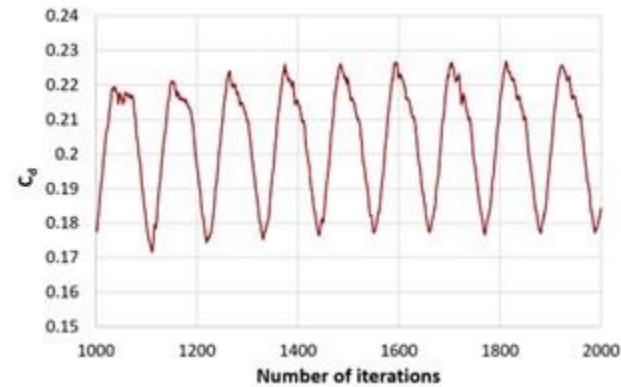
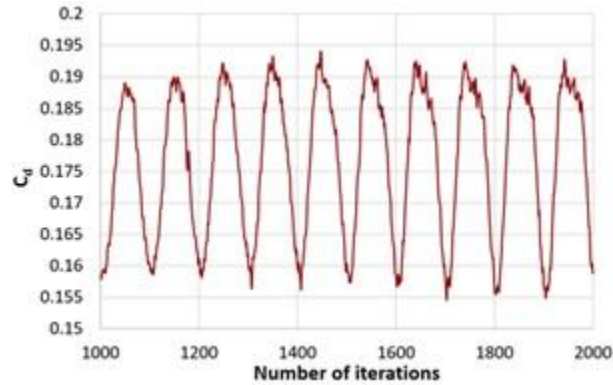
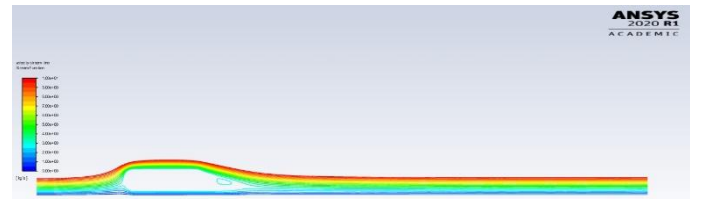


Figure: Graph of residuals

Figure: Static pressure contours (slant angle  $205^\circ$ )

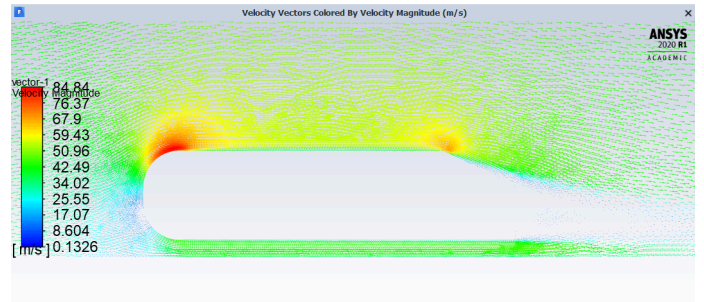
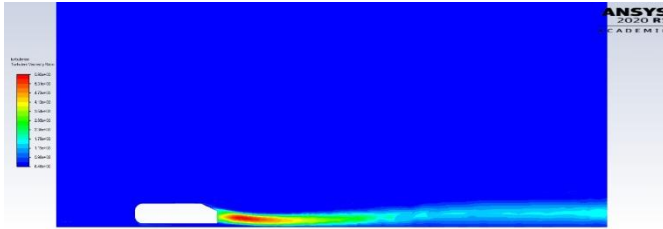


Figure: Stream function contours (slant angle  $205^\circ$ )



Comment: Vortices were observed after the body, since the flow was turbulent.

Figure: Turbulent viscosity ratio contours (slant angle  $205^\circ$ )



Comment: maximum velocity vectors is 84 m/s which at sharp of the geometry.

Figure: Pressure coefficient over the Ahmed body (slant angle  $205^\circ$ )

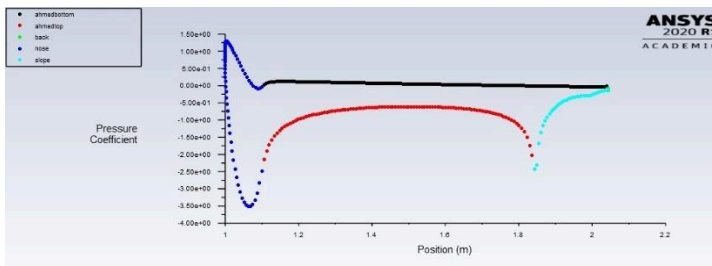


Figure: drag coefficient with time step (slant angle  $205^\circ$ )

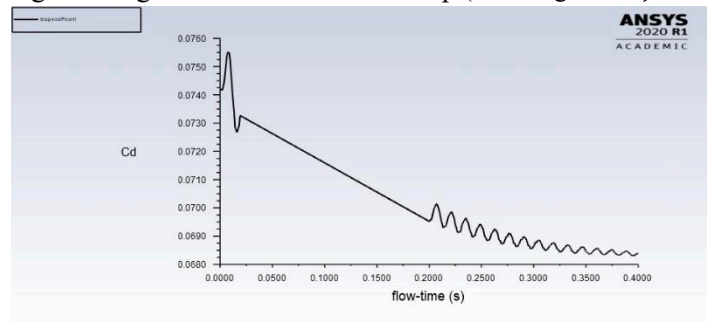


Figure: Turbulence over the Ahmed body (slant angle  $205^\circ$ )

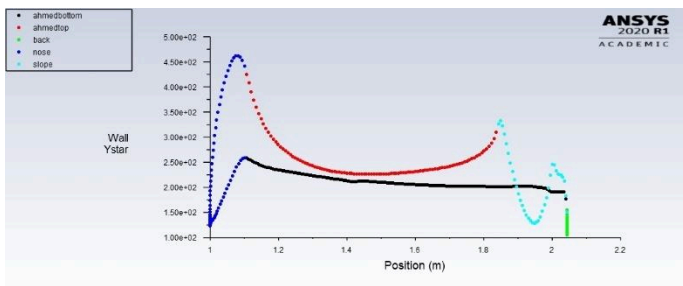


Figure: Wall Fluxes skin friction coefficient over the Ahmed body (slant angle  $205^\circ$ )

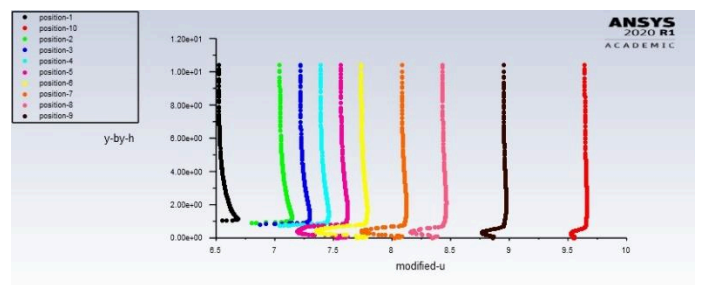
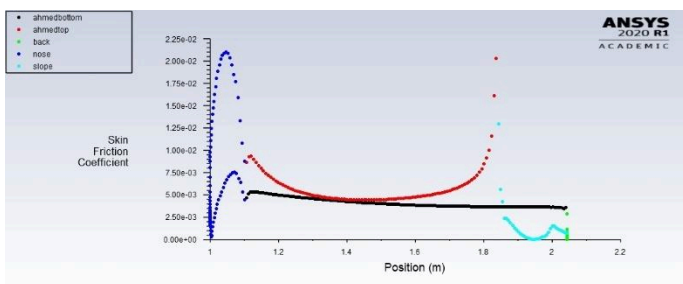


Figure: y-by-h vs Modified-U function (slant angle  $205^\circ$ )

Figure: Velocity vectors colored by velocity magnitude (m/s) (slant angle  $205^\circ$ )

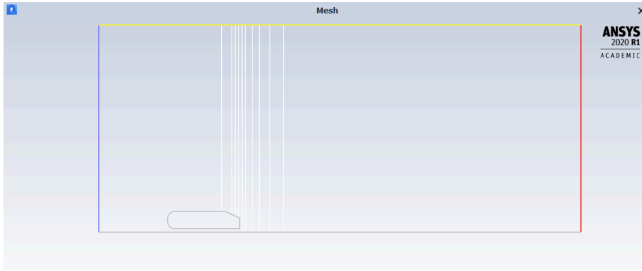


Figure: Creating lines to plot modified TKE and modified  $U(\text{slant angle } 205^\circ)$

| Surface Name | x0      | y0   | x1      | y1 |
|--------------|---------|------|---------|----|
| position-1   | 1.78192 | 0.05 | 1.78192 | 3  |
| position-2   | 1.932   | 0.05 | 1.932   | 3  |
| position-3   | 1.98208 | 0.05 | 1.98208 | 3  |
| position-4   | 2.03191 | 0.05 | 2.03191 | 3  |
| position-5   | 2.08201 | 0    | 2.08201 | 3  |
| position-6   | 2.13212 | 0    | 2.13212 | 3  |
| position-7   | 2.23206 | 0    | 2.23206 | 3  |
| position-8   | 2.332   | 0    | 2.332   | 3  |
| position-9   | 2.482   | 0    | 2.482   | 3  |
| position-10  | 2.6819  | 0    | 2.6819  | 3  |

Figure: Line position table (slant angle  $205^\circ$ )

Animation of stream function: (slant angle  $205^\circ$ )

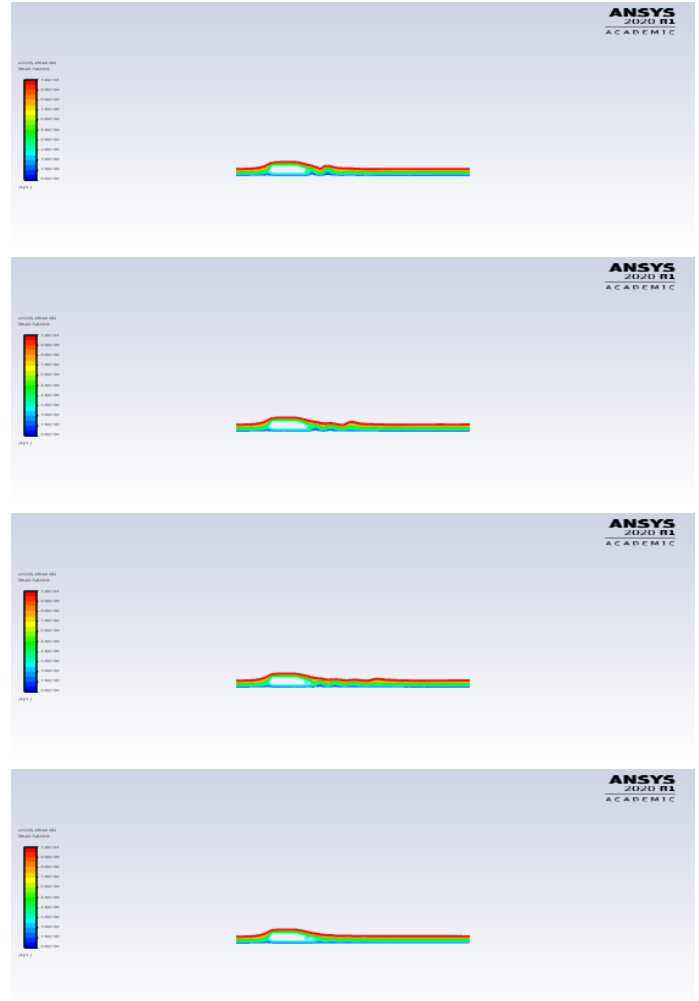
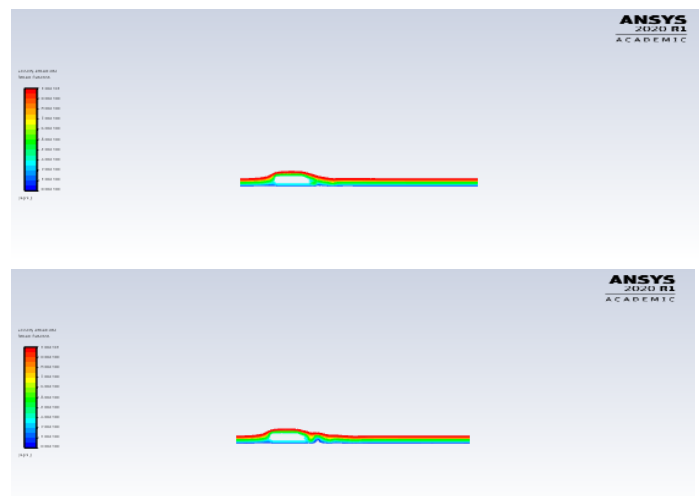


Figure: Animation of stream function

## CONCLUSION

CFD simulation has been carried out to investigate the flow characteristics over a model car (Ahmed Body). Turbulent flow is expected in the model since the first speed is given high. In the turbulent flow, As the fluid particles increase in speed, the flow begins to become irregular. In this case, there are grain transitions between the flowing layers. Flow lines intersect, the particles blend together to form a turbulent flow. The drag coefficient is found around 0.0695. The drag coefficient of an automobile are found by changing the slant angle of the Ahmed body. The drag coefficient was found to increase with increase in the slant angles due to the development of vortex region at the downstream side of the Ahmed body.

## REFERENCES

- [1] <https://www.researchgate.net/publication/318722228>
- [2] <http://www.ijettjournal.org>
- [3] Engineeringtoolbox, AeroSpace, Mechanical & Mechatronic Eng.  
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